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Milestone 2: ML Solution DataViz	
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Program and Specialization	BSIT - Data Analytics

I. Project Overview

The purpose of creating these visualizations is to transform initial analysis and examine it graphically to form insights from the cleaned dataset from FinMark. The previous EDA described the data mainly through tables of counts and descriptive statistics, this time, it visualizes a solution that shows the shape of the data, how values are distributed, how variables relate to one another, how activity changes over time, and where unusual observations sit. While data can be analyzed through summary statistics, it can hide structure that can lead to more insights. Two variables can share the same mean and spread yet look very different once plotted, and issues such as spikes created by imputation, repeated identifiers, or class imbalance are far easier to see in a visualized chart or cross-tabulation than in numbers.

The objective of this solution data visualization is to generate statistics and visualizations, analyze and interpret distributions, relationships, trends, and anomalies, and cross-check the visual results against the EDA findings, and lastly to document the insights relevant to FinMark's problem of turning inconsistent customer data into actionable information. This supports future machine learning activities by showing which variables are balanced and ready for modeling, which need special handling—for example, Sentiment and the zero-value transactions—and which relationships are too weak to be useful predictors. These findings directly inform feature selection, segmentation, and the choice of modeling approach moving forward.

II. Visualization Overview

Figure	Visualization	Chart type	Variables
1	Customer Ages and Transaction Amounts Distributions	Histogram with KDE	Age, Amount
2	Customer Age and Transaction Amount Boxplots	Boxplot	Age, Amount
3	Distribution of Social Media Interaction Types	Count plot	InteractionType
4-5	Transaction Volume by Product Category and Payment Method	Horizontal bar	ProductCategory, PaymentMethod
6	Customer Age vs. Average Transaction Amount	Scatter with regression line	Age, average Amount
7-8	Average Amount and Transactions per Customer by Age Group	Bar chart	AgeGroup, Amount, TransactionCount
9-10	Average Transaction Amount by Product Category and Payment Method	Horizontal bar	ProductCategory, PaymentMethod, Amount
11-12	Monthly Transaction Counts and Total Transaction Amount	Line chart	TransactionDate, counts, totals
13	Monthly Social Media Interaction Counts	Line chart	InteractionDate, counts
14	Monthly Social Media Interactions vs. Transaction Counts	Scatter with regression line	Monthly interactions, monthly transactions
15	Sentiment Distribution by Social Media Platform	100% stacked bar	Platform, Sentiment
16	Interaction Types by Social Media Platform	100% stacked bar	Platform, InteractionType
17	Sentiment Distribution by	100% stacked bar	InteractionType,

	Interaction Type		Sentiment
18	Zero-Value vs. Non-Zero Transactions	Bar chart	Amount (flagged)

***Table 1**. Summary of visualizations developed*

A. Visualization Objectives

The visualizations generated were intended to show findings about:

- How Age and Amount are distributed and whether any outliers remain after cleaning
- How the categorical variables (Product Category, Payment Method, Platform, Interaction Type, Sentiment) divide up the records
- Whether customer characteristics such as Age relate to spending behavior
- How transactions and social media activity move over time and relative to each other.

The explored trends, comparisons, and relationships between variables were Age vs. Average Transaction Amount (both continuous and by age group), Average Amount and Transaction Volume by Product Category and Payment Method, Monthly Transaction Counts and totals across the 24 months, Monthly Social Media Interactions across the 12 months, the relationship between Monthly Interactions and Monthly Transactions, and how Sentiment varies by Platform and Interaction Type. This analysis prioritized Age and Amount (the only numerical variables), Sentiment (the one imbalanced categorical variable identified previously), and the date variables that allow the three datasets to be compared over time. The datasets were linked through the shared CustomerID key where a customer-level view was needed. These visualizations were prioritized as they matter to the project, as they test whether the cleaned data

can support the intended machine learning applications and identify the problems that must be handled first.

B. Visualization Tools and Workflow

In Google Colab, Pandas was used for loading, grouping, merging, and cross-tabulation, NumPy for statistics, Matplotlib for figure layout, and Seaborn for statistical plots, with the whitegrid theme applied for readability. NumPy functions produced the descriptive statistics, the quartile and IQR outlier fences, the Pearson correlation coefficients, and the flag separating zero-value from non-zero transactions. Seaborn's histplot, boxplot, barplot, countplot, and regplot generated the main figures, while Pandas' plot was used for the percentage-stacked comparisons. The cleaned datasets were re-verified to ensure zero duplicate rows and zero missing values. The workflow then computed numerical statistics, plotted distributions, built customer-level and month-level aggregates, and visualized relationships. Several preprocessing steps and numerical outputs supported the charts: the median-imputed Age and Amount values, the Undetermined categories, a per-customer aggregation of average amount and transaction count (1,871 customers), an age-group binning (18-25 through 66-70), monthly resampling by month start, and a check of repeated CustomerIDs in the demographics data.

III. Visualization Review and Analysis

A. Distribution and Frequency Analysis

Statistic	*Customer Age*	*Transaction Amount*
Count	*3,023*	*3,015*
Mean	*44.69*	*493.62*
Median	*45.00*	*498.01*
Standard deviation	*14.42*	*279.25*

Variance (population)	*207.98*	*77,983.20*
Minimum / Maximum	*18.00 / 70.00*	*0.00 / 999.86*
Q1 / Q3	*34.00 / 57.00*	*257.31 / 720.39*
IQR	*23.00*	*463.08*
Outlier fences (lower / upper)	*-0.50 / 91.50*	*-437.31 / 1,415.01*
Potential outliers	*0*	*0*

Table 2. *NumPy descriptive statistics and IQR outlier check*

The NumPy statistics as shown above in Table 2 reproduce the previous EDA results. Age has a mean of 44.69, a median of 45.00, a standard deviation of 14.42, and a range of 18 to 70. Amount has a mean of 493.62, a median of 498.01, a standard deviation of 279.25, and a range of 0.00 to 999.86. The quartiles show the shape of each: the middle half of customers is between 34 and 57 years old, with the median sitting almost exactly at the center of that interval. Transaction amounts have Q1 = 257.31, Q2 = 498.01, and Q3 = 720.39, which are close to the quartiles of an even spread across 0 to 1,000 (250, 500, 750). Transaction amounts are therefore not concentrated at low or high values but are distributed broadly across the full range.

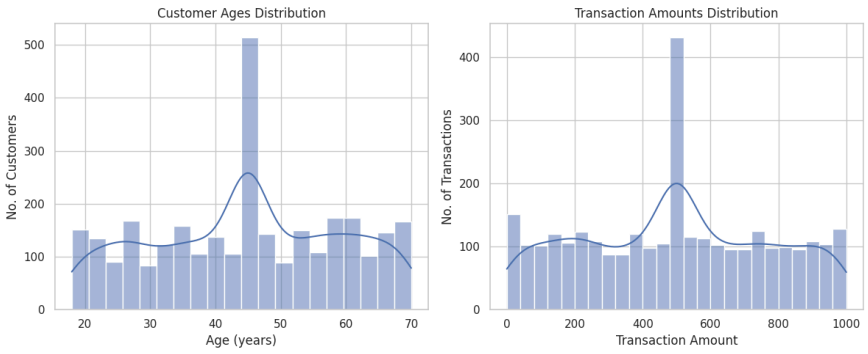


Figure 1. *Customer Ages and Transaction Amounts Distributions (histogram with KDE)*

Figure 1 shows the shape of both variables. The Age histogram is centered in the mid-40s with one very tall bar at 45 (roughly 510 customers, compared with about 80 to 170 in the other bars). The Amount histogram is broad and fairly flat across 0 to 1,000, with one very tall bar near 498 (roughly 430 transactions, compared with about 90 to 125 elsewhere). Both spikes come from the median imputation performed in the EDA (median age of 45 and median amount of 498.015), not from natural customer behavior. The first Amount bar is also slightly elevated at 150, because the 60 zero-value transactions fall in it.

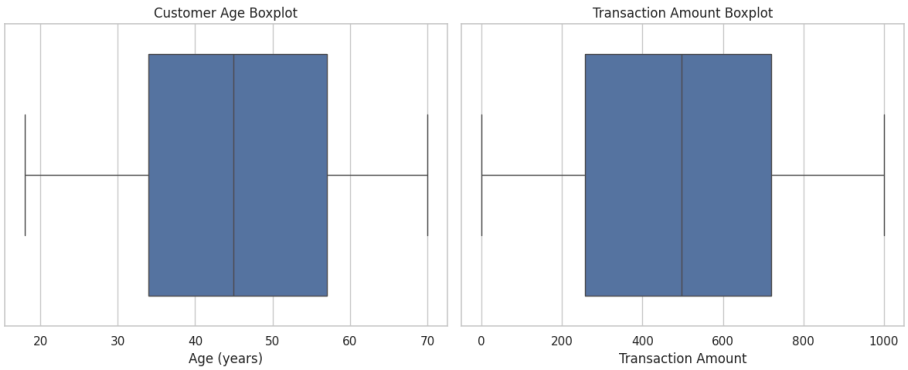


Figure 2. *Customer Age and Transaction Amount Boxplots*

The boxplots in Figure 2 agree with the IQR calculations. The boxes are centered on the medians, the whiskers run the full observed range (18 to 70 and 0 to about 1,000), and no points are plotted beyond them.

Variable	Distribution
Interaction type (n = 3,020)	○ Comment - 1,018 (33.7%) ○ Share - 1,007 (33.3%) ○ Like - 995 (33.0%)
Product category (n = 3,000)	○ Clothing - 568 (18.9%) ○ Electronics - 561 (18.7%) ○ Automotive - 538 (17.9%)

	○ Home & Garden - 529 (17.6%) ○ Health & Beauty - 524 (17.5%) ○ Undetermined - 280 (9.3%)
Payment method (n = 3,000)	○ Debit Card - 762 (25.4%) ○ Credit Card - 753 (25.1%) ○ PayPal - 747 (24.9%) ○ Bank Transfer - 738 (24.6%)
Sentiment (n = 3,020)	○ Positive - 905 (30.0%) ○ Neutral - 874 (28.9%) ○ Negative - 866 (28.7%) ○ Undetermined - 309 (10.2%) ○ Very Negative - 34 (1.1%) ○ Very Positive - 32 (1.1%)

Table 3. *Categorical frequency summary*

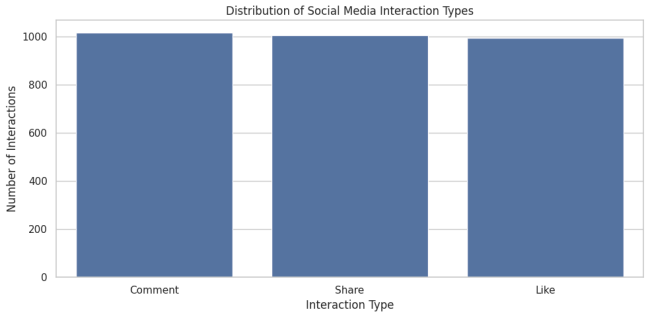


Figure 3. *Distribution of Social Media Interaction Types*

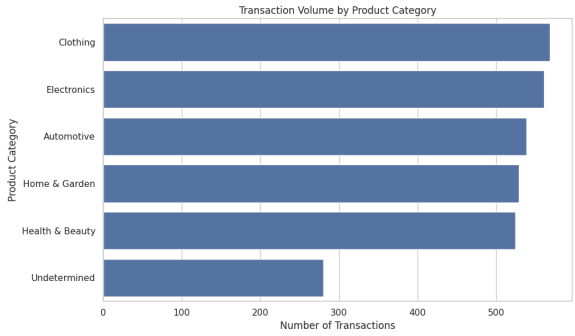


Figure 4. *Transaction Volume by Product Category*

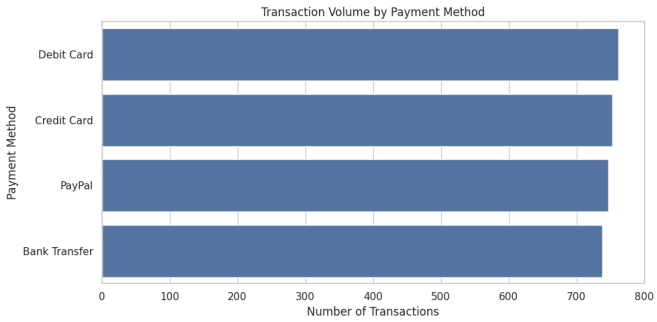


Figure 5. *Transaction Volume by Payment Meth*

The categorical charts confirm that most variables are close to evenly divided in Table 3. Interaction types are almost equal (Figure 3), and product categories and payment methods are similarly balanced (Figures 4 and 5). Clothing leads Product Category with 568 unique transactions, and Debit Card leads Payment Method with 762, but no group dominates. Sentiment is the clear exception: Positive, Neutral, and Negative each account for about 29%, while Very Negative and Very Positive together account for about 2.2% of interactions, and 10.2% are Undetermined.

The distributions reveal that the numerical variables are bounded and moderately spread, the categorical variables are balanced, which is favorable for modeling, the imputed values appear as artificial peaks in the numerical histograms, and the extreme sentiment classes are scarce and will need special handling. The Undetermined category forms a visible bar of its own (about half the height of the named categories in Figure 4) and should be kept as an explicit level rather than dropped.

B. Relationship and Trend Analysis

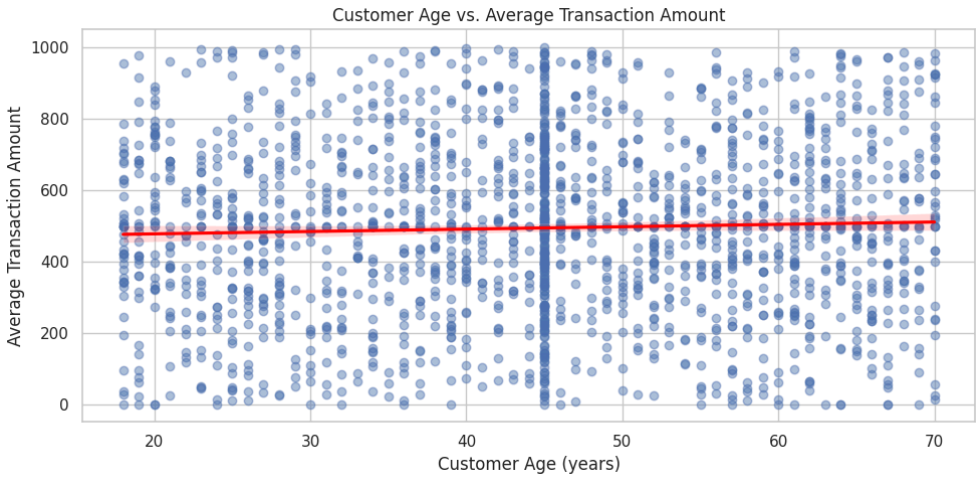


Figure 6. *Customer Age vs. Average Transaction Amount*

Age group	Customers	Avg. amount	Avg. transactions per customer	Total amount
18-25	246	468.95	1.60	186,982.42
26-35	280	474.15	1.58	211,884.72
36-45	531	505.17	1.65	441,515.30
46-55	307	506.15	1.64	253,555.12
56-65	345	491.99	1.56	266,612.52
66-70	153	504.15	1.50	117,922.05

Table 4. *Spending summary by age group (customers with transactions, excluding conflicting ages)*

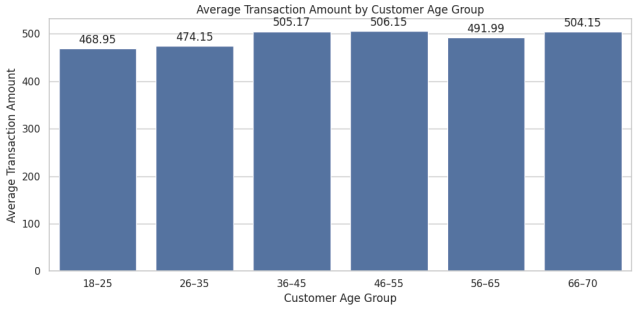


Figure 7. *Average Transaction Amount by Customer Age Group*

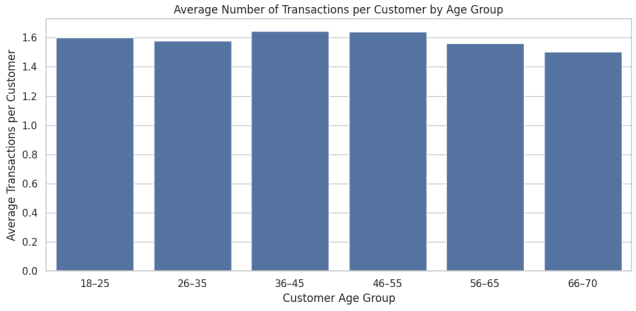


Figure 8. *Average Transactions per Customer by Age Group*

The scatter plot of Age against average transaction amount (Figure 6) shows a wide, shapeless cloud of points with a nearly flat regression line that rises only from about 480 at age 18 to about 510 at age 70. The correlation coefficient is 0.039, which is negligible. A dense vertical band at age 45 is visible in the plot, it is the imputed age again. The age-group bar charts (Figures 7 and 8, Table 4) agree that average transaction amount ranges only from 468.95 (18-25) to 506.15 (46-55), and average transactions per customer range from 1.50 (66-70) to 1.65 (36-45). Customers aged 36 and above spend slightly more per

transaction (about 492 to 506) than customers aged 18 to 35 (about 469 to 474), but the differences are small relative to the standard deviation of about 279. The 36-45 group generates the largest total amount (441,515.30) mainly because it contains the most customers (531), followed by the 56-65 group (266,612.52) and the 46-55 group (253,555.12)

Group	Avg. amount	Transactions	Total amount
Home & Garden	*515.46*	*529*	*273,194.31*
Automotive	*500.23*	*538*	*270,126.48*
Clothing	*494.23*	*568*	*282,204.26*
Electronics	*487.45*	*561*	*274,920.35*
Undetermined	*479.67*	*280*	*135,267.44*
Health & Beauty	*478.33*	*524*	*252,555.67*
Credit Card	*499.87*	*753*	*377,398.56*
PayPal	*496.49*	*747*	*372,368.27*
Debit Card	*495.22*	*762*	*379,830.49*
Bank Transfer	*482.73*	*738*	*358,671.19*

Table 5. *Transaction spending by product category and payment method*

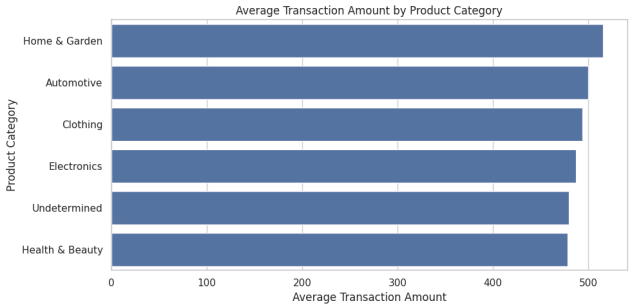


Figure 9*. Average Transaction Amount by Product Category*

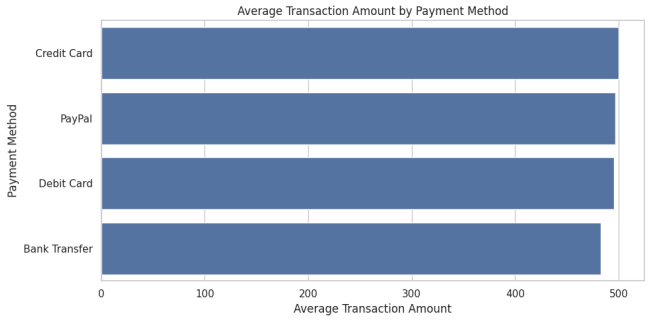


Figure 10. *Average Transaction Amount by Payment Method*

Average amount per transaction is similar across categories (Figure 9, Table 5). Home & Garden is highest (515.46), followed by Automotive (500.23), Clothing (494.23), and Electronics (487.45), and Health & Beauty is lowest (478.33), a gap of roughly 7%. Clothing has the highest transaction volume (568) and the largest total amount (282,204.26), while Home & Garden has the highest average but fewer transactions (529). Payment methods are similarly close (Figure 10): Credit Card has the highest average (499.87), and Bank Transfer has the lowest average (482.73) and the lowest volume (738). The Undetermined category (average 479.67) behaves like the named categories, suggesting the missing categories did not hide a distinct group.

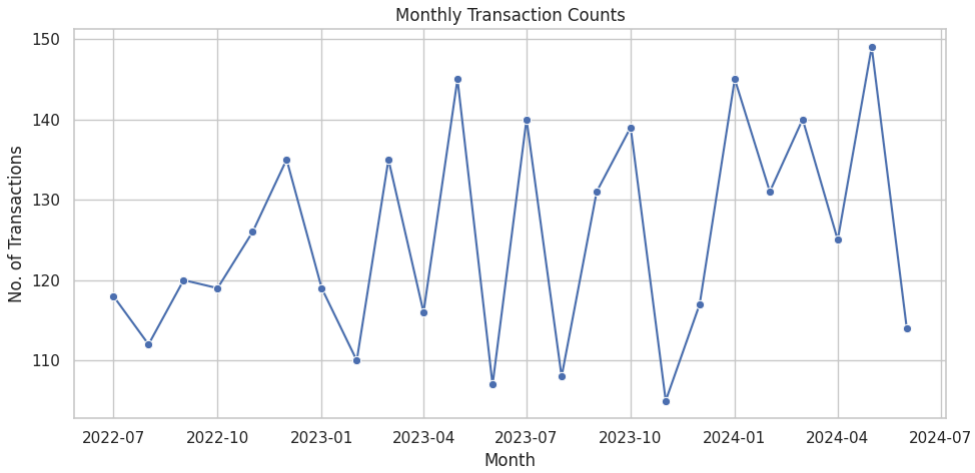


Figure 11. *Monthly Transaction Counts*

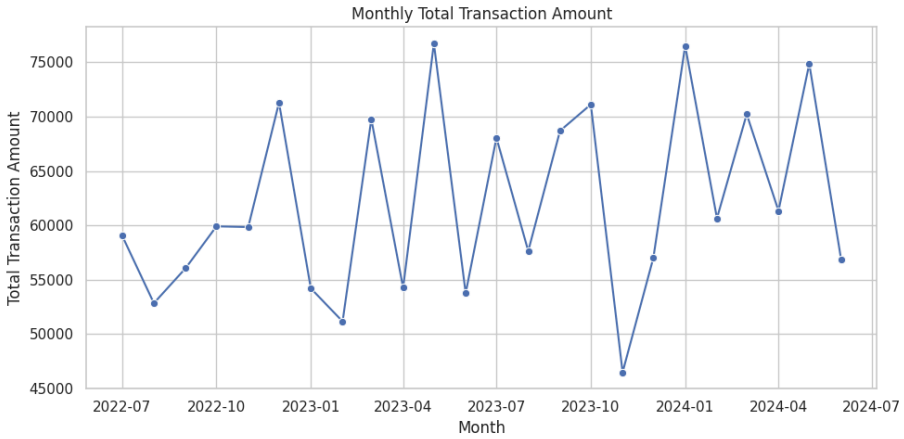


Figure 12. *Monthly Total Transaction Amount*

Monthly transaction counts (Figure 11) fluctuate widely from month to month, from 105 to 149 transactions, without a smooth trend. The series zigzags between high and low months, most sharply from mid-2023. The first twelve months (July 2022 to June 2023) averaged about 122 transactions per month, while the second twelve (July 2023 to June 2024) averaged about 129, a modest increase of roughly 6%. Total monthly amount (Figure 12) follows transaction count closely, because the monthly average amount stays between about 443 and 534 (Table 6, July 2023 to June 2024). The highest totals occur in May 2023 (about 76,700), January 2024 (76,522.00), and May 2024 (74,856.44), and the lowest is November 2023 (46,475.22 | 105 transactions). The pattern is not consistently seasonal. December 2022 was a peak (135 transactions) but December 2023 was below average (117), and November reversed in the same way (126 in 2022, 105 in 2023). Only a few months behave alike in both years, March and May are high in both (135 and 140, 145 and 149), while June and August are low in both (107 and 114, 112 and 108).

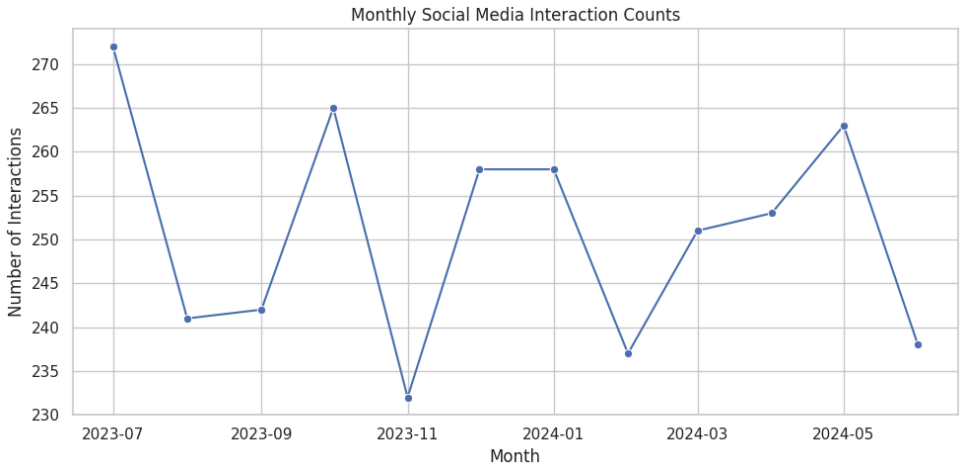


Figure 13. *Monthly Social Media Interaction Counts*

Month	Transactions	Total amount	Avg. amount	Interactions
2023-07	140	68,052.50	486.09	272
2023-08	108	57,632.70	533.64	241
2023-09	131	68,693.11	516.49	242
2023-10	139	71,117.02	507.98	265
2023-11	105	46,475.22	442.62	232
2023-12	117	57,009.76	483.13	258
2024-01	145	76,522.00	527.74	258
2024-02	131	60,646.57	462.95	237
2024-03	140	70,222.59	501.59	251
2024-04	125	61,330.09	490.64	253
2024-05	149	74,856.44	502.39	263
2024-06	114	56,886.88	499.01	238

Table 6. *Monthly transactions and social media interactions, July 2023 to June 2024*

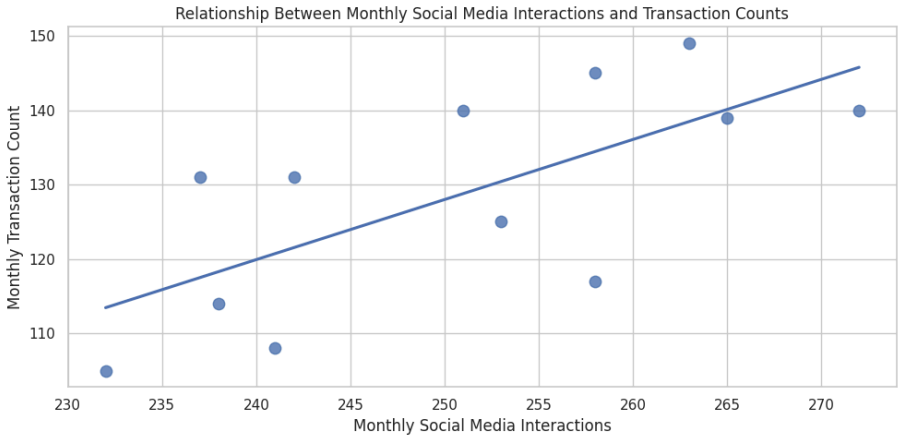


Figure 14. *Relationship Between Monthly Social Media Interactions and Transaction Counts*

Monthly social media interactions (Figure 13) are more stable than transactions, ranging from 232 (November 2023) to 272 (July 2023), an average of about 251 per month. Their coefficient of variation is about 5%, compared with about 11% for monthly transactions. Over the 12 months the two series share (July 2023 to June 2024, Table 6), the correlation between monthly transaction count and monthly interaction count is 0.696, a moderately strong positive relationship (Figure 14). Interactions peak in July 2023 and transactions in May 2024; both series are lowest in November 2023. The scatter also shows considerable spread around the line, with months of 258 interactions having 117 and 145 transactions. Because the relationship rests on only 12 monthly data points, it should be treated as an indicator for further analysis rather than evidence that social activity drives purchases.

Platform	Interactions	Positive rate	Negative rate
Facebook	891	30.19%	27.61%
Instagram	921	31.49%	29.42%
Twitter	917	26.72%	29.66%

Table 7. *Positive and negative sentiment rates by platform (Undetermined platform is excluded)*

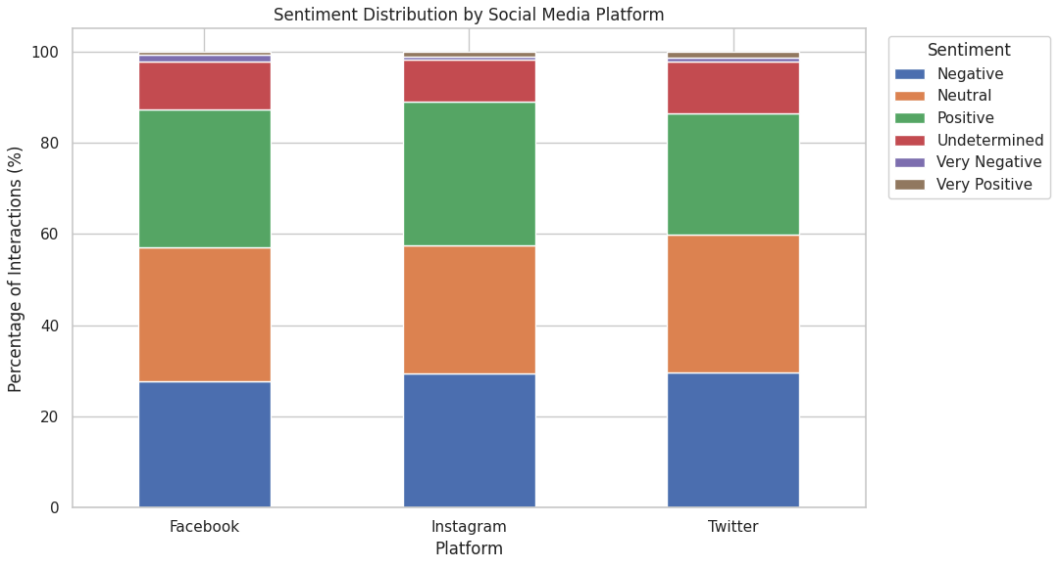


Figure 15. *Sentiment Distribution by Social Media Platform*

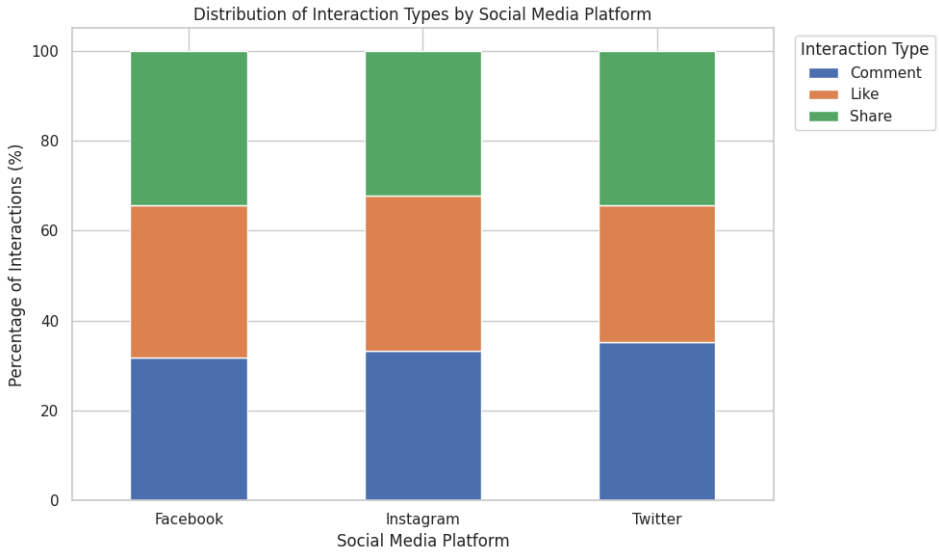


Figure 16. *Distribution of Interaction Types by Social Media Platform*

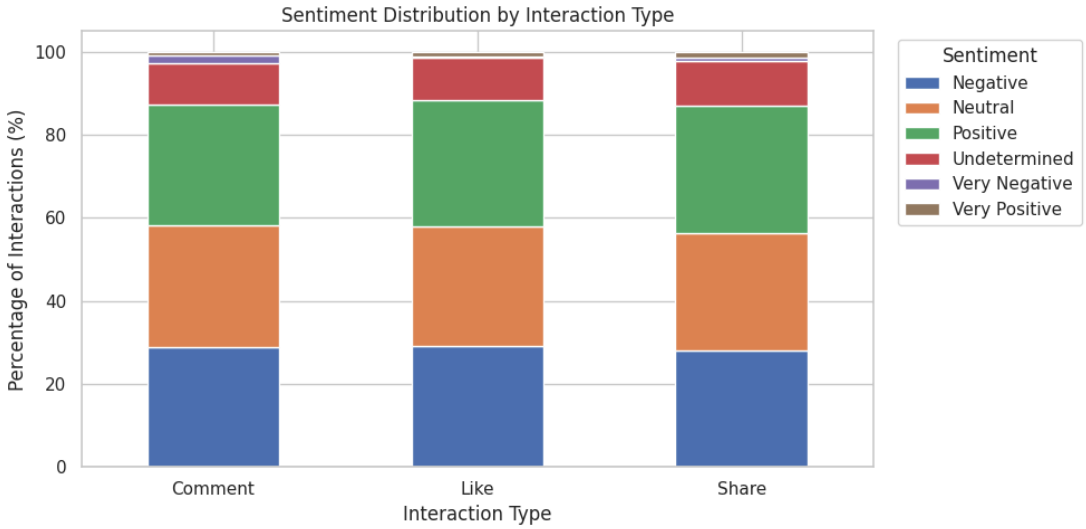


Figure 17. *Sentiment Distribution by Interaction Type*

The platform charts show nearly identical interaction-type mixes (Figure 16). Each type makes up roughly 30% to 35% of a platform's interactions. Sentiment differs slightly by platform (Figure 15, Table 7). Instagram has the highest share of Positive interactions (31.49%) and Twitter the lowest (26.72%). Twitter is the only platform where Negative (29.66%) is higher than Positive; Facebook is 30.19% Positive and 27.61% Negative, and Instagram is 31.49% Positive and 29.42% Negative. The Positive rate varies by about five percentage points across platforms and the Negative rate by about two, so the differences are modest. By interaction type (Figure 17), the moderate classes are almost identical, but Very Negative sentiment is concentrated in comments. Of the 34 Very Negative interactions, 21 are comments (2.06% of all comments), compared with 3 likes (0.30%) and 10 shares (0.99%). The counts are small, so this is a signal to monitor rather than a firm conclusion.

The weak age-spending correlation indicates that age alone is unlikely to be a strong predictor of transaction amount, which suggests that segmentation may need behavioral variables rather than demographics alone. The balanced

categorical variables are suitable as features without resampling, and the monthly relationship between social activity and purchasing suggests a time-aware feature that can be tested further.

C. Outliers and Anomalies

The boxplots (Figure 2) and the IQR calculations (Table 2) agree that there are no statistical outliers. Age has fences of -0.50 and 91.50, and Amount has fences of -437.31 and 1,415.01, both variables have a potential outlier count of 0, and the observed ranges (18-70 and 0-999.86) lie well inside the fences. This confirms that the removal of the invalid ages (-1 and 150) and the invalid amount (-100) in Milestone 1 was effective.

The most visible irregularities are not outliers but spikes created by median imputation, the concentration of ages at 45 (Figures 1 and 6) and the concentration of amounts near 498.01 (Figure 1). Both inflate the central bar of their histograms, slightly understate true variability, and make the 36-45 age group (531 customers in Table 4) appear much larger than its neighboring groups (280 and 307).

Potential anomalies:

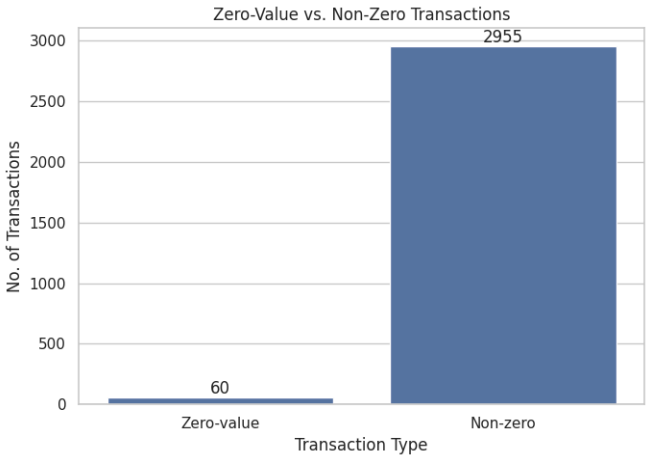


Figure 18. *Zero-Value vs. Non-Zero Transactions*

- There are 60 transactions with an Amount of 0 (1.99% of all transactions, Figure 18). They are spread across categories and payment methods. They are not flagged by the IQR rule because the lower fence is negative, so only a deliberate check reveals them. Likely causes are promotional, complimentary, or trial transactions (the "Free" entries from the raw data). Counts within single groups, such as Bank Transfer (9) and Automotive (7), are too small to interpret on their own.

- The demographics file contains 3,000 unique CustomerIDs across 3,023 rows. 23 IDs appear on more than one row, and 11 of them have different ages recorded. In every one of these conflicting pairs, one of the two ages is 45, consistent with an imputed age sitting next to a real value. Merging demographics with transactions without handling this produced 1,889 matched rows for 1,871 unique customers.

- This is why the category and payment-method counts based on unique IDs (summing to 3,000) differ slightly from the Milestone 1 row counts (summing to 3,015). The Interaction type and Sentiment counts in Table 3 remain row-based (3,020).

- Social media data exists only for July 2023 to June 2024, while transactions cover July 2022 to June 2024.

Imputation spikes can bias distance-based and distribution-sensitive models and reduce the apparent variability of Age and Amount. Zero-value transactions can pull down averages and mislead revenue models if treated as ordinary purchases. Repeated customer and transaction identifiers can cause row inflation when joining datasets and

can leak information between training and test sets if the same customer or transaction is split across them. These observations matter because model performance depends on whether the data represents real customer behavior rather than artifacts of cleaning.

IV. Relationship Between Visualizations and Milestone 1 Findings

The visualizations supported nearly all of the Milestone 1 observations, the symmetric Age distribution, the broad Amount distribution, the absence of outliers, the balance of the categorical variables, the imbalance of Sentiment, the 60 zero-value transactions, and the repeat-activity pattern of roughly 1.6 transactions per customer. The NumPy results match the Milestone 1 values for the mean, median, standard deviation, minimum, maximum, and range. The Amount median appears as 498.01 in NumPy output and 498.02 in Milestone 1 because the imputed value is 498.015 and is rounded differently. Neither difference changes any conclusion.

The visualizations also added several findings. The median-imputation spikes at Age 45 and Amount 498.01 are clearly visible in the histograms and the scatter plot, and they explain why the 36-45 group is unusually large. Age has almost no relationship with spending. Monthly transactions and monthly social media interactions are positively related. Transaction volume swings month to month with no consistent seasonality and only a mild year-over-year increase. Twitter is the only platform with more Negative than Positive interactions, and Very Negative sentiment is concentrated in comments. In addition, 11 customers have conflicting ages, and 15 TransactionIDs and 20 InteractionIDs are repeated, none of which were identified in Milestone 1. Finally, spending differences across categories, payment methods, and age groups are small (under 8%).

V. Visualization Insights and Interpretation

The cleaned dataset from FinMark is internally consistent and balanced across nearly all categorical variables. Transaction amounts are spread broadly between 0 and about 1,000 with an average near 494, and customers transact about 1.6 times on average. The analysis also identified issues that cleaning alone did not resolve, imputation artifacts, zero-value transactions, repeated identifiers, and the scarcity of extreme sentiment labels. Spending is remarkably uniform across age groups, product categories, and payment methods, with differences of under 8%. Monthly transaction activity moves within a band of roughly 105 to 149 transactions with a mild upward drift, and monthly social media activity is more stable. Where transactions are higher, social media interactions also tend to be higher.

Age is not a meaningful standalone predictor of spending. The strongest relationship found is the monthly link between social media interactions and transaction counts, which suggests that engagement metrics may be useful features for time-based modeling. The 36–45 age group is the largest segment and generates the most total spending though partly because of the age-45 imputation.

Potential issues should be accounted for and considered, median-imputed values create artificial peaks, and the roughly 9% to 10% Undetermined share in each imputed categorical field remains a real category. Zero value transactions can distort revenue models. Repeated CustomerIDs, TransactionIDs, and InteractionIDs can inflate joins and cause data leakage. Extreme sentiment classes (about 1% each) are too rare to model without adjustment, and the platform and interaction-type differences in sentiment are small and based on small counts.

FinMark's challenge is converting inconsistent data into actionable insight. The visualizations show that the data is now reliable enough for exploratory customer analysis, but that simple demographic splits, such as age, will not explain spending on their own. For FinMark, customer behavior (frequency, category mix, engagement) is more likely to separate customers than demographics. Social media engagement appears connected to purchasing activity at the monthly level, which supports

FinMark's interest in combining the three data sources, and negative comments on Twitter and in comment threads are worth monitoring for FinMark's brand management.

VI. Project Resources and Supporting Files

Reference No.	Reference	Link
1	Final Cleaned Dataset	GDrive | Github
2	Python Code for Exploratory Data Analysis	Github
3	Python Code for Data Visualizations	Github
4	Generated Visualization Outputs	Github